Detecting Network Scanning Activity Using Nmap and Wireshark

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Background

1. Problem and importance
    - Network scanning identifies open ports and services on a device. Administrators use it to check systems, but can be used by attackers to scan and gather information before attempting an intrusion. A key aspect of this project is whether a defender can recognize common Nmap scan behavior by examining captured TCP traffic. The goal is to find a set of packet indicators that distinguish controlled scans from normal connections in a lab.
2. Challenges
    - Legit programs can open several network connections, which can cause false positives.
    - Different scan types produce different TCP patterns.
    - Open and closed ports can produce different responses.
    - A small virtual lab may not accurately represent real-world networks.
3. Objectives
    - Compare normal TCP traffic with Nmap TCP connect scans and SYN scans.
    - Identify scanning indicators such as destination ports, SYN patterns, connection completion, resets, and timing.
    - Create simple detection rules to identify possible network scans.
    - Test how accurately the rules detect scans and how often they flag normal traffic.

Proposed Methodology

1. Possible solutions
    - I'll set up a controlled virtual network with a scanning machine and a target that I am authorized to test, and capture the traffic in Wireshark. First I'll generate normal connections. Then I'll run Nmap TCP connect (-sT) and SYN (-sS) scans and save the packet captures along with the Nmap commands and output so I can compare them. For detection, I'll start with a few simple rules such as many distinct destination ports contacted in a short time, repeated SYN attempts, and incomplete versus completed TCP handshakes. Once I've looked at the captures, I'll adjust the rules to fit the traffic I actually see.
2. Evaluation
    - The platform is Nmap, Wireshark, and isolated virtual machines. A small script or spreadsheet can count packets or connections if that turns out to be useful. I'll run each scan type more than once, record the settings for every run, and repeat the captures. For each known scan and baseline capture, I'll record whether a rule triggers. The report will cover detections, misses, and false positives, and a short comparison of how TCP connect and SYN scans behave. The analysis will be descriptive.

Expected contributions

Outcomes

- Packet captures of normal traffic, TCP connect scans, and SYN scans.

- A comparison of the different traffic patterns.

- A table showing useful indicators and their limitations.

Contributions

- A repeatable experiment showing which TCP indicators are useful for detecting scans.

- Identify when detection rules may produce false positives.

- Focus on comparing and evaluating existing techniques rather than creating a new algorithm.

Impacts

- Help beginners understand how network scanning appears in packet captures.

- Show why detection rules need to be tested against legitimate network traffic.

Targeted conference/journal

The International Symposium on Research in Attacks, Intrusions and Defenses(RAID), is a relevant venue for related research. It focuses on intrusion detection and defensive security, which fits the scan-detection experiment I'm planning. https://raid2026.org/call.html.

Project management

1. Team
    a. I plan on completing this project by myself, I have experience with networking, wireshark, and Nmap through previous class assignments and projects. During this project I will have to improve my skills that involve looking over detections and calculating accuracy as I have limited experience in that. To keep the scope manageable I will focus on two nmap scan types using a virtual environment.
2. Research plan
    a.

Month	Research goal
Sept	Proposal, website
Oct	Research, method design, report 1 (Oct 31
Nov	Lab setup, capture/analysis, report 2 (Nov 14)
Dec	Evaluation, presentation (Dec 12), final paper (Dec 13)

References

University groups
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